

Module 4 :: Minimization of Switching Functions

Practice Problems

1. With the aid of a four-variable Karnaugh map, derive minimal sum-of-products expressions for each of the following functions:

(a) $f_1(w, x, y, z) = \sum(0, 1, 2, 3, 4, 6, 8, 9, 10, 11);$

(b) $f_2(w, x, y, z) = \sum(0, 1, 5, 7, 8, 10, 14, 15);$

(c) $f_3(w, x, y, z) = \sum(0, 2, 4, 5, 6, 8, 10, 12).$

2. Find the minimal sum-of-products and minimal product-of-sums expressions for:

$$f(w, x, y, z) = \prod(1, 4, 5, 6, 11, 12, 13, 14, 15).$$

3. Let $f = \sum(5, 6, 13)$ and $f_1 = \sum(0, 1, 2, 3, 5, 6, 8, 9, 10, 11, 13)$. Find f_2 such that $f = f_1 \cdot f_2'$.

4. Use the map method to generate the set of prime implicants and obtain all minimal expressions for the following functions:

(a) $f_1(w, x, y, z) = \sum(1, 5, 6, 12, 13, 14) + \sum_{\phi}(2, 4)$

(b) $f_2(v, w, x, y, z) = \sum(0, 1, 3, 8, 9, 13, 14, 15, 16, 17, 19, 24, 25, 27, 31)$

(c) $f_3(w, x, y, z) = \sum(0, 1, 4, 5, 6, 7, 9, 11, 15) + \sum_{\phi}(10, 14)$

(d) $f_4(v, w, x, y, z) = \sum(1, 5, 6, 7, 9, 13, 14, 15, 17, 18, 19, 21, 22, 23, 25, 29, 30)$

(e) $f_5(w, x, y, z) = \sum(0, 1, 5, 7, 8, 10, 14, 15)$